Inspire the love of learning

WO7-2353235

Q1. Genetic Algorithm

A binary chromosome select features. Its fitness is:

$$F(z) = Acc_{val}(z) - 0.02|z|$$

where $|z|$ is the number of selected features. Let $A = 101100$ with accuracy 0.86 and $B = 111110$ with accuracy 0.89. Compute fitness, cross after bit 3, and flip bit 6 of the first child. Why use validation accuracy?

$F(A) = 0.86 - 0.02|3| = 0.8$

$F(B) = 0.89 - 0.02|5| = 0.79$

=> A has higher fitness despite lower accuracy.

Crossover:

$101 | 100, \quad 111 | 110$

$\Rightarrow \quad 101110, \quad 111100$

Mutation: $101110 \rightarrow 101111$ (5 features).

Validation guides search; repeatly using test accuracy would leak test information into model selection.

Q2. Bayesian Network Factorization

For the DAG:

$$A \rightarrow C \leftarrow B, \quad C \rightarrow D, \quad B \rightarrow E$$

factorize the joint distribution. If all variables are binary, count independent parameters.

$$P(A,B,C,D,E) = P(A)P(B)P(C|A,B)P(D|C)P(E|B)$$

A binary node with k binary parents needs 2^k independent parameters => $1+1+4+2+2 = 10$

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Q3. D-Separation

Does the DAG above imply each independence? Explain the effect of observing a common effect.

- $A \perp B$: Yes. Collider C blocks $A \rightarrow C \leftarrow B$
- $A \perp B \mid C$: No. Observing the collider actives the path
- $A \perp B \mid D$: No. Observing a descendant of collider C also activates the path.
- $A \perp D \mid C$: Yes. Observing the middle node blocks the chain $A \rightarrow C \rightarrow D$.

Observing a common effect can couple independent causes: evidence for one cause may reduce the need for another.

Q4. Naive Bayes vs. TAN

For class Y and features X_1, X_2, X_3, compare NB with TAN containing $X_1 \rightarrow X_2 \rightarrow X_3$ and Y as a parent of every feature. Give a HAR example & one trade-off.

NB: $P(Y, X_1, X_2, X_3) = P(Y)\,P(X_1 \mid Y)\,P(X_2, Y)\,P(X_3, Y)$

TAN: $P(Y, X_1, X_2, X_3) = P(Y)\,P(X_1 \mid Y)\,P(X_2 \mid Y, X_1)\,P(X_3 \mid Y, X_2)$

NB assume all features are conditionally independent given Y. TAN allows 1 additional feature parent per feature in a tree.

In HAR, acceleration magnitude & acceleration energy may remain related within an activity class. NB can double-count their evidence, producing overconfident probabilities. TAN captures some dependencies, but requires more parameters & structure learning. Blueangel